

COMET PROTOTYPE

The prototype of the de Havilland Comet made its maiden flight in 1949. The first jet airliner to enter service, the Comet suffered a spate of crashes caused by metal fatigue.



arrived on the scene in 1950. It proved an immensely popular aircraft, offering a smoother, quieter ride than any piston-engined airliner. Although it only carried 50 passengers, it was run at a profit by British European Airways, which in the 1950s was responsible for a quarter of all passenger traffic in Western Europe. Britain went on to make the larger Bristol Britannia and Vickers Vanguard turboprops, while America belatedly came up with the Lockheed turboprop Model 188 Electra, an aircraft whose reputation was severely damaged by a series of shocking crashes.

The country that produced the most remarkable of all turboprop airliners was the Soviet Union. The Tupolev Tu-114 was a spin-off from Soviet military aviation, developing out of the Tu-95 long-range turboprop bomber. Entering service with Aeroflot in 1957, the Tu-114 was a giant – the largest airliner in the world before the Boeing 747. It could carry 170 passengers over 5,500 miles (8,800km) between stops, cruising at around 480mph (770kph). In 1959 a Tu-114 flew from Moscow to New York in 11 hours, and in the 1960s the Tupolevs operated scheduled services linking the Soviet capital to Cuba, Canada, West Africa, India, and Japan.

de Havilland's Comet

In the course of the 1950s, however, it became obvious that the future lay not in hybrid turboprops, but in pure turbojets. Here, too, it was Britain that took the first step. The de Havilland Comet, which began flight testing in

for the limited number of tickets – the aircraft initially only seated 36 – and Comets were soon being flown by other airlines, including Air France. The American business magazine *Fortune* declared 1953 “the year of the Coronation and the Comet.”

Disaster struck in January 1954, when a Comet flying out of Rome exploded at 30,000ft (9,000m). An earlier Comet crash in India had been blamed on bad weather, but this time there seemed no obvious explanation. Flights were resumed, but three months later another Comet disintegrated over the Mediterranean. While experts sought a cause for these disasters, all Comets were grounded. Lengthy investigation finally

revealed that the aircraft had a structural weakness around the windows. A redesigned Comet resumed commercial flights four years later, stretched to carry 72 passengers. It was just in time to make BOAC the first airline to operate a scheduled jet service across the Atlantic (with one stop at Gander), but by then the technological and commercial lead had been lost.

During the Comet's period out of service, the only jet airliner operating anywhere in the world was the Soviet Tu-104, another product of Andrei Tupolev's design bureau and another derivative of the Cold War nuclear bomber program. From 1956 onward it successfully established a number of scheduled services, including Moscow to Beijing via Irkutsk, and had an excellent safety

record. The main beneficiary of the Comet debacle, however, was the American aircraft industry, which grasped the opportunity to catch up with and far surpass the foreign competition.

Boeing breakthrough

Until the 1950s Boeing had no great reputation as a manufacturer of commercial aircraft. Its strength was large military airplanes. But this provided the pathway to the aircraft that truly founded the era of the passenger jet: the Boeing 707. While Douglas, the world's leading manufacturer of civilian passenger aircraft, reflected the skepticism of most American airlines about the commercial feasibility of jet air travel, Boeing was already heavily involved in the development of large military jet aircraft such as the B-47 and B-52 bombers. Boeing boss Bill Allen figured that if he could make a jet airplane that would be saleable both to the Air Force and to civilian airlines, he would be able to make a profit.

In 1954 Boeing rolled out the prototype designated 367-80 and universally known as the “Dash 80.” It was pitched to the USAF as primarily an in-flight refueling aircraft to service the new jet bomber fleet, a role it eventually fulfilled under the military designation KC-135. For airlines, Boeing found it had to offer a significantly

“At jet speed you could circle the globe in 40 hours. The world was shrinking to half the size.”

HAROLD MANSFIELD

VISION: THE STORY OF BOEING



TURBOFAN 707

Introduced into airline service in 1958, the Boeing 707 was the aircraft that made jet travel a commercial success. This 707 prominently displays the fact that it is equipped with turbofan engines, which became standard on the airliner from 1962.